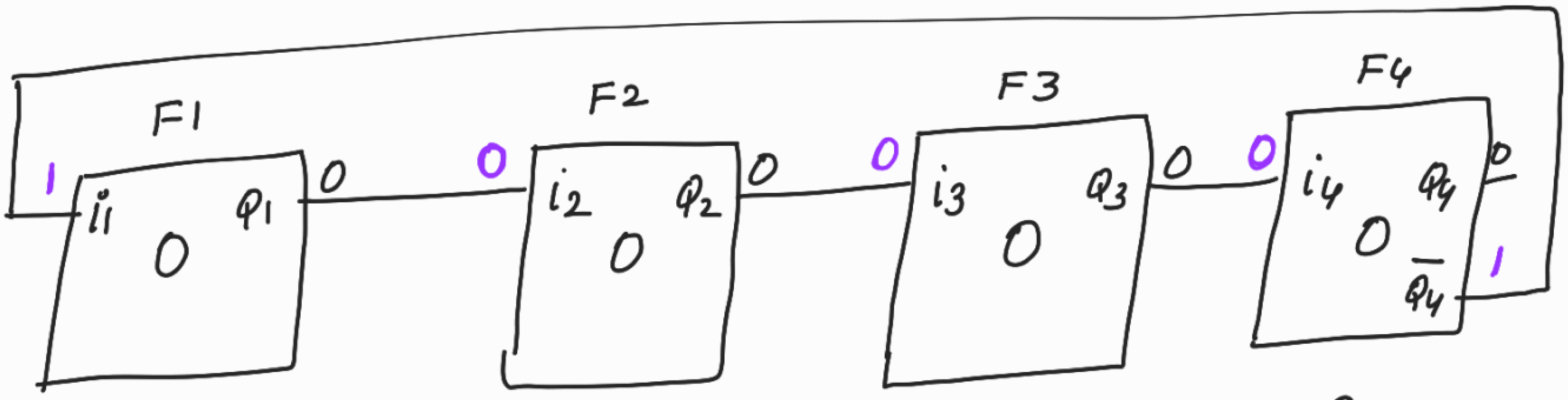
DOMAIN MIXING :-

Keeping ffs with different scan clks in the same scan chain.

NOTE : (JUST FOR EXPLANATION)



	Q1	Q2	Q3	Q4
initial	0	0	0	0
↑	1	0	0	0

↳ 1st rising edge

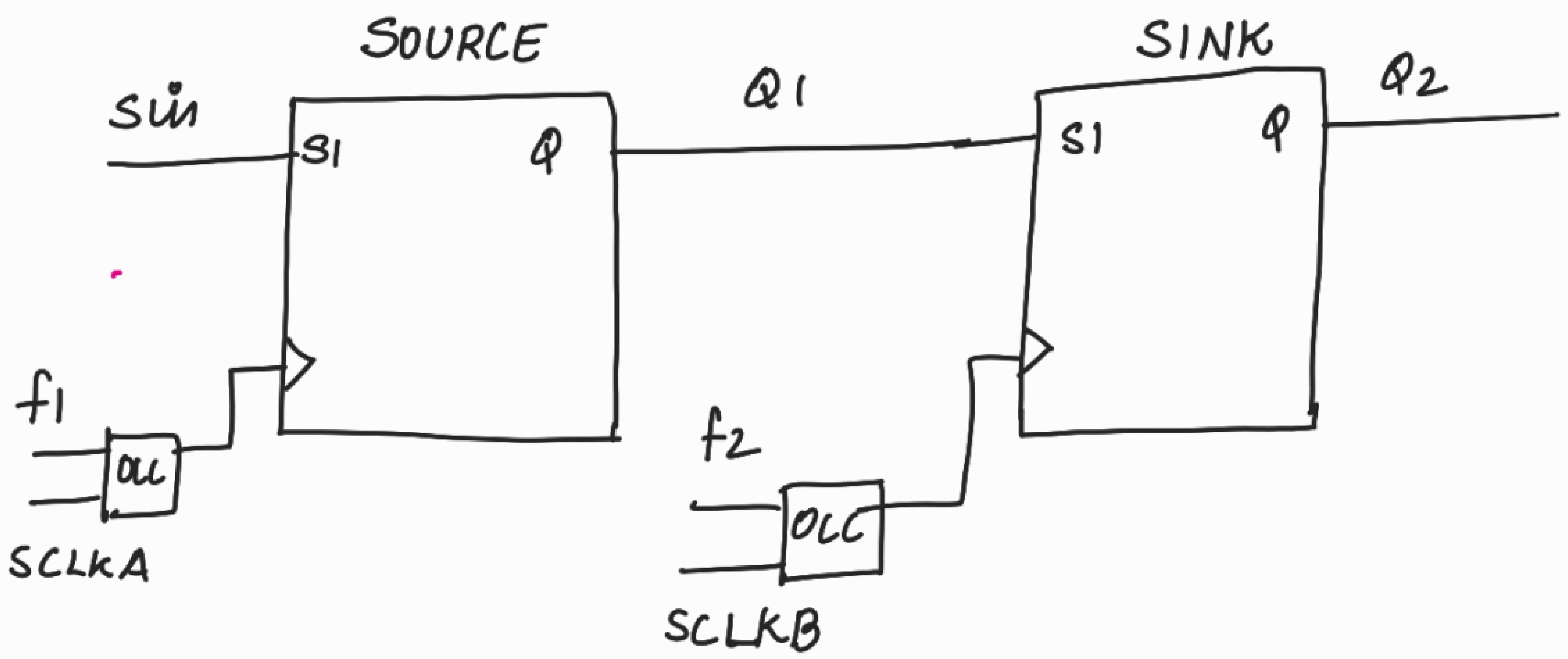
↳ previous value of Q_1 has gone into F2

This is a normal operation.

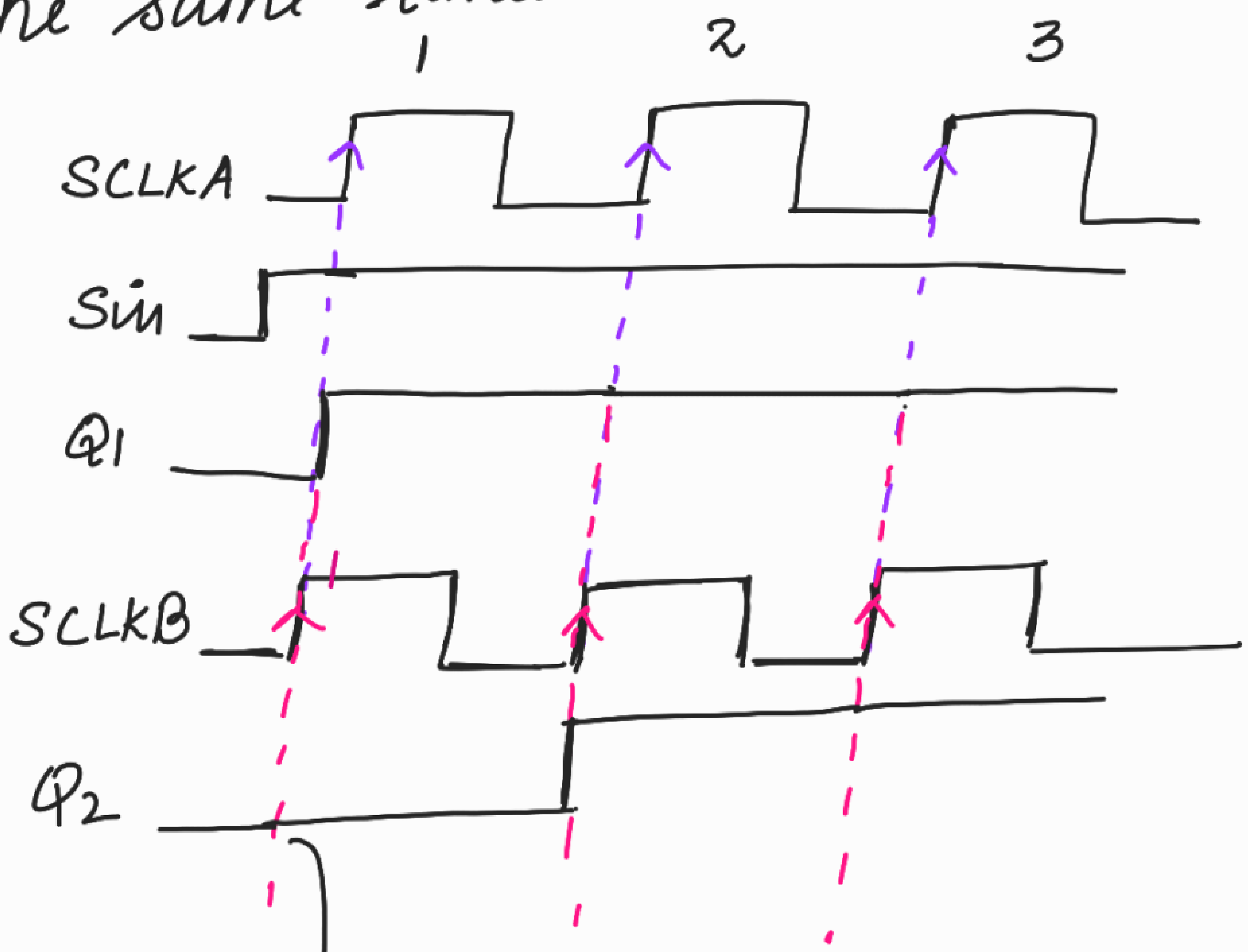
In this eg, all the flops are having the same clk.

Even when the flops are having diff clocks, the same principle has to be followed.

PROBLEM IN DOMAIN MIXING :-



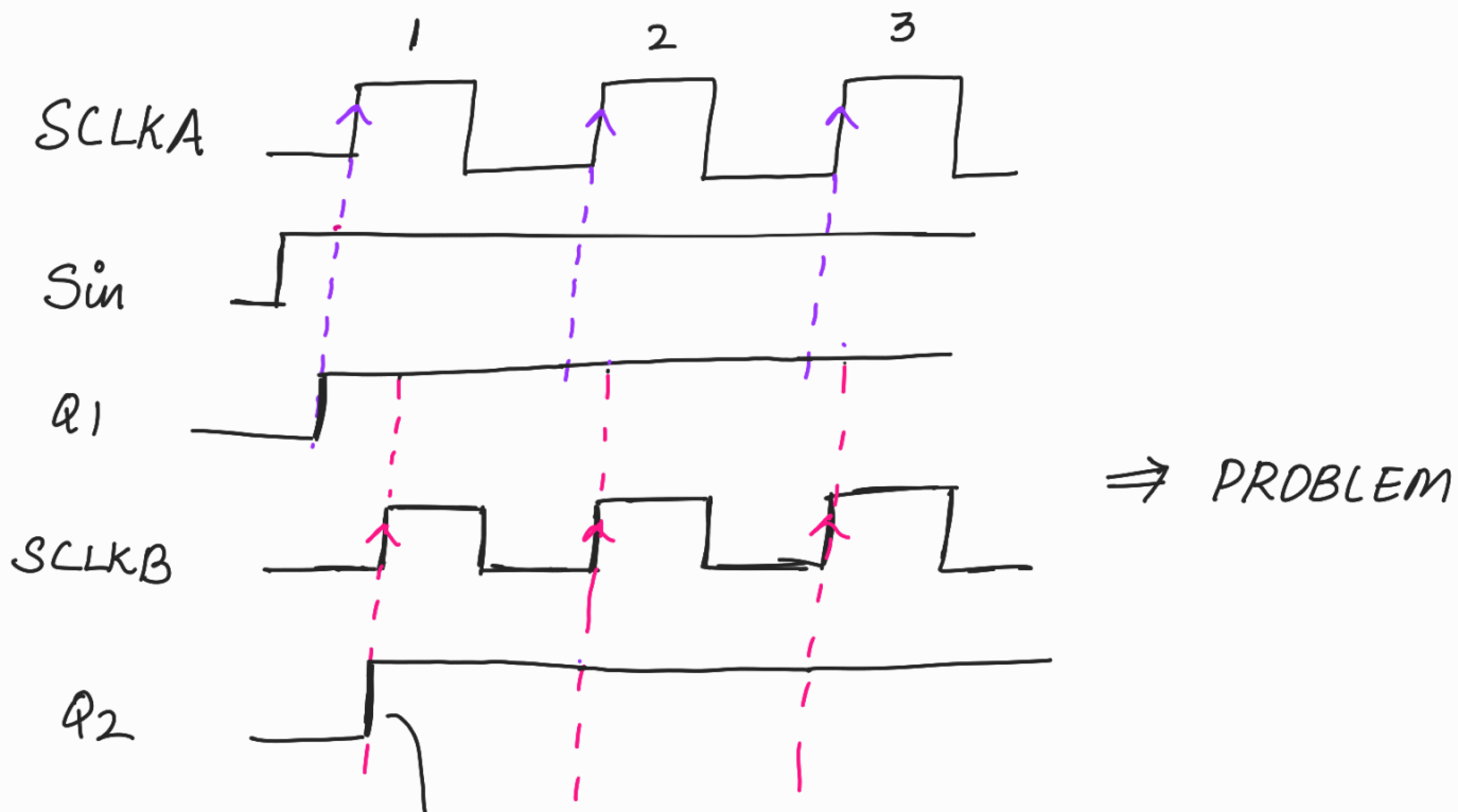
Consider that SCLKA and SCLKB are arriving at the same time.



⇒ NOT A PROBLEM because it follows the same principle as discussed above.

at this edge (1st rising edge of its clk), $Q_2 = 0$ (which is the previous value of Q1)

Consider that SCLKB occurs little later than SCLKA



at this edge itself Q2 has become 1 (which is the current value of Q1)

Actually Q2 should have been 0 at this edge (previous value of Q1)

When 2 adjacent flops (source and sink) in a scan path are clocked by different shift clocks (scan clks), the sink must not take in data from the source at the time the source changes its value.

↳ Eg:- source's o/p has changed to 1 at the 1st rising edge of its clk.

Sink's o/p should not change to 1 at the 1st rising edge of its clk itself.

But when clk skew exists b/w shift clks, this rule fails.

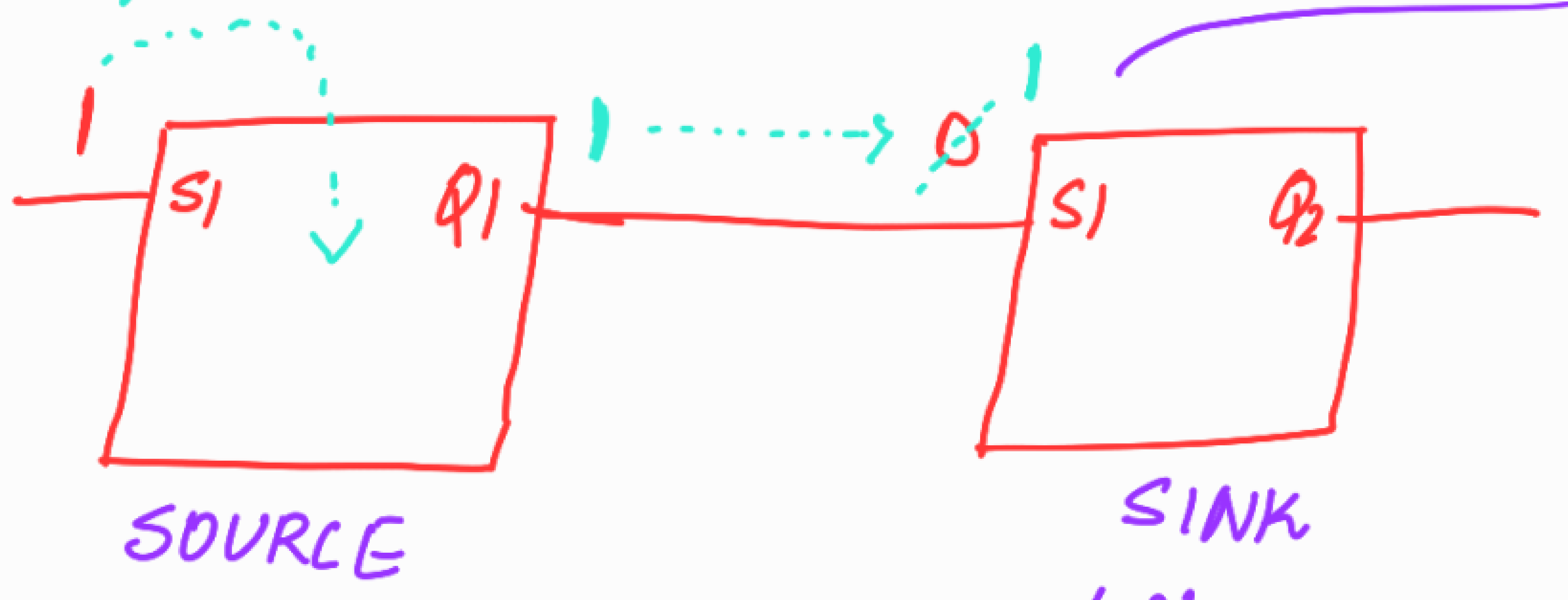
i.e; sink's o/p also changes to 1 at the 1st rising edge of its clk.

Failure to satisfy this rule can result in unwanted shoot-through during scan chain shifting when clk skew exists b/w the diff shift clocks (scan clocks)

See *

* New value replaces the old value.

whenever rising edge of SCLKA comes,

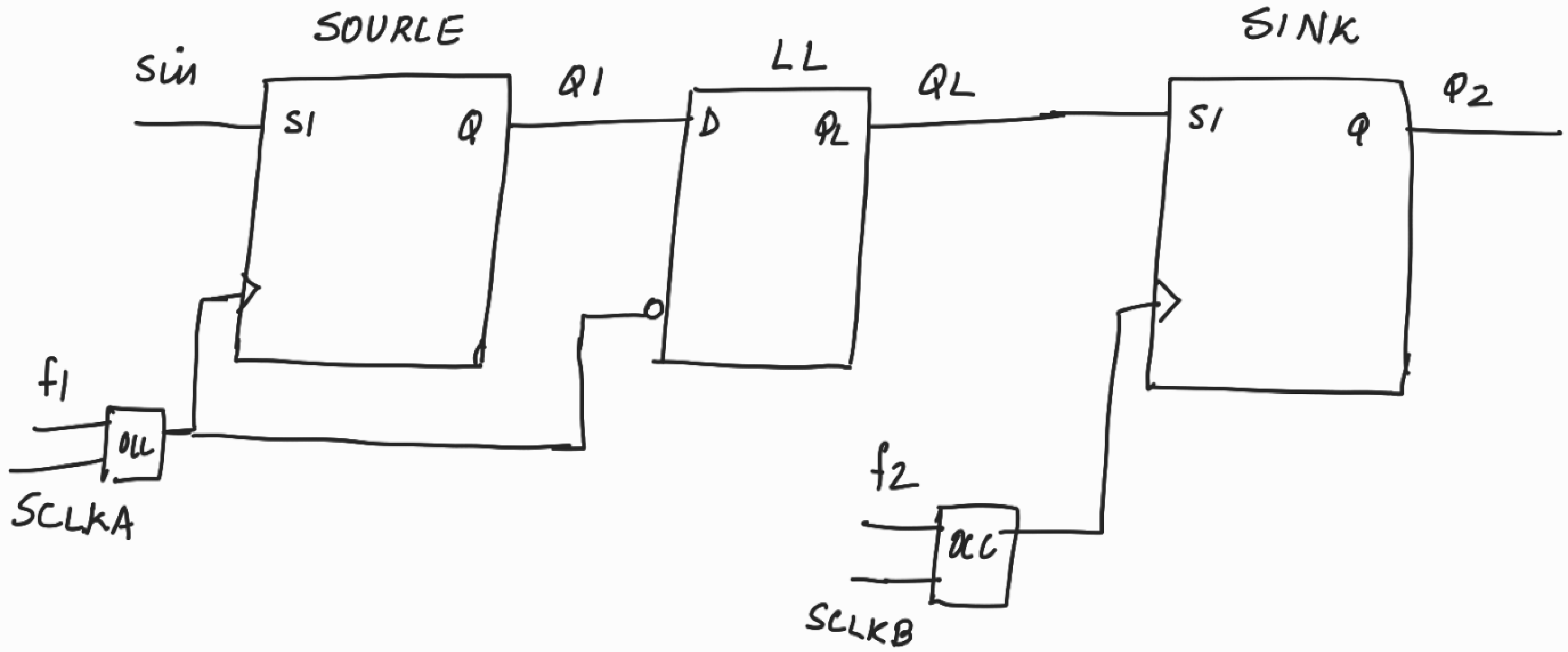


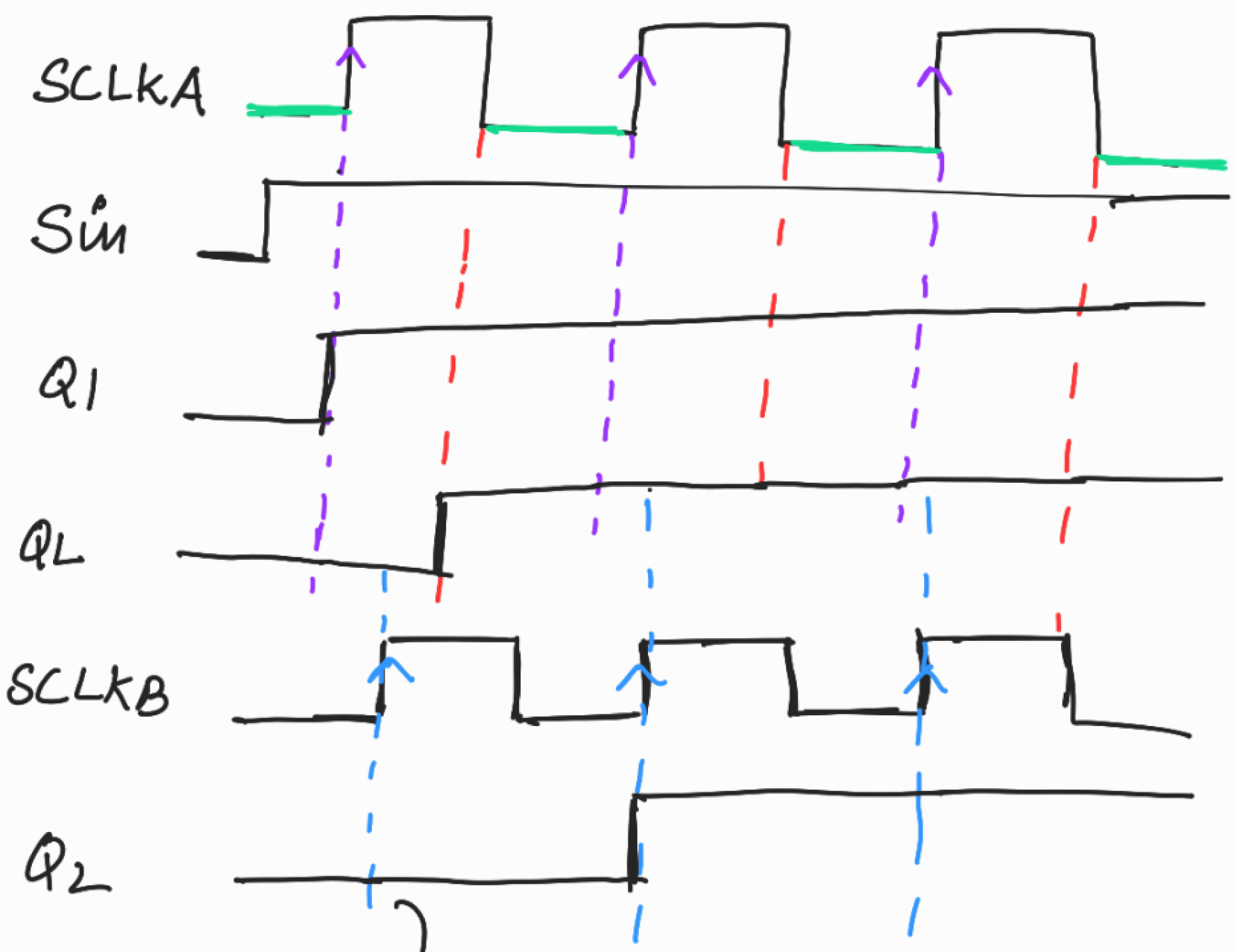
All this has happened by the time the rising edge of SCLKB comes.

When rising edge of SCLKB comes

Q2 = 1.

Fix :-





⇒ PROBLEM IS SOLVED.

$Q_2 = 0$ (which is the previous of Q1)

NOTE :- The skews of clock will be known only after doing STA (which is in turn done after PD).

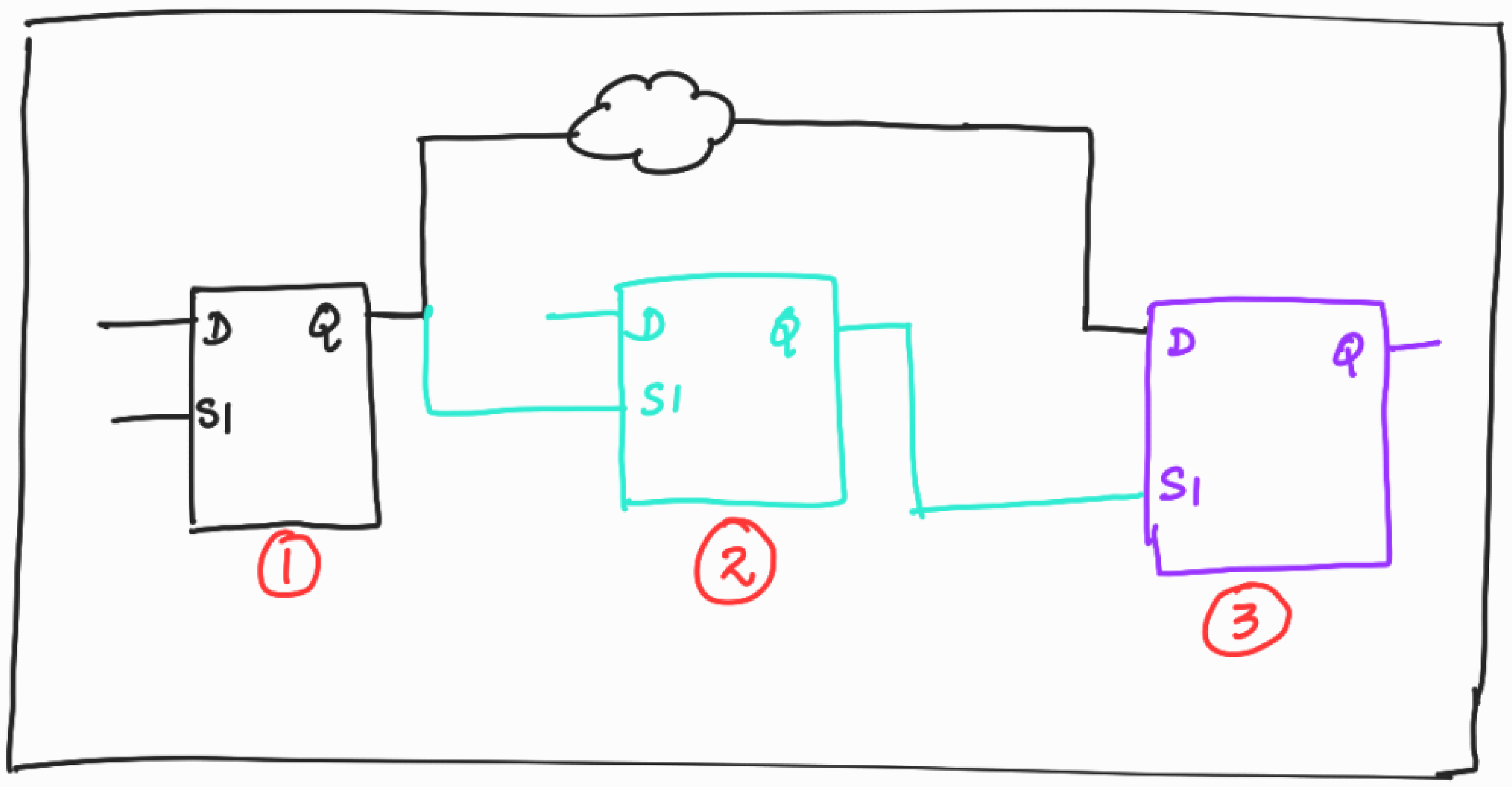
During DFT, our tool doesn't know the skews of clock.

∴ Our tool will add a lock up latch wherever there is domain mixing.

———— x ————

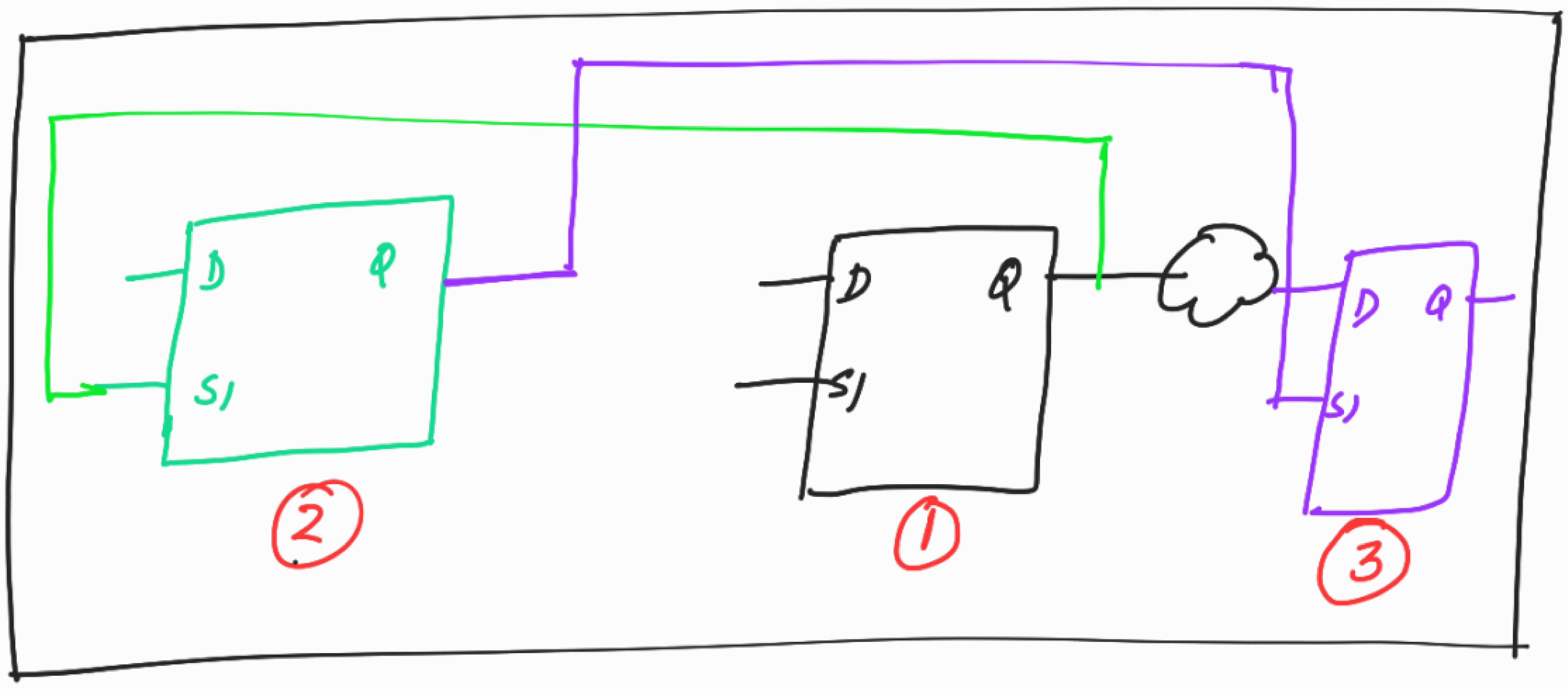
SCAN CHAIN RE-ORDERING :-

This is done by PD team



⇒ After DFT

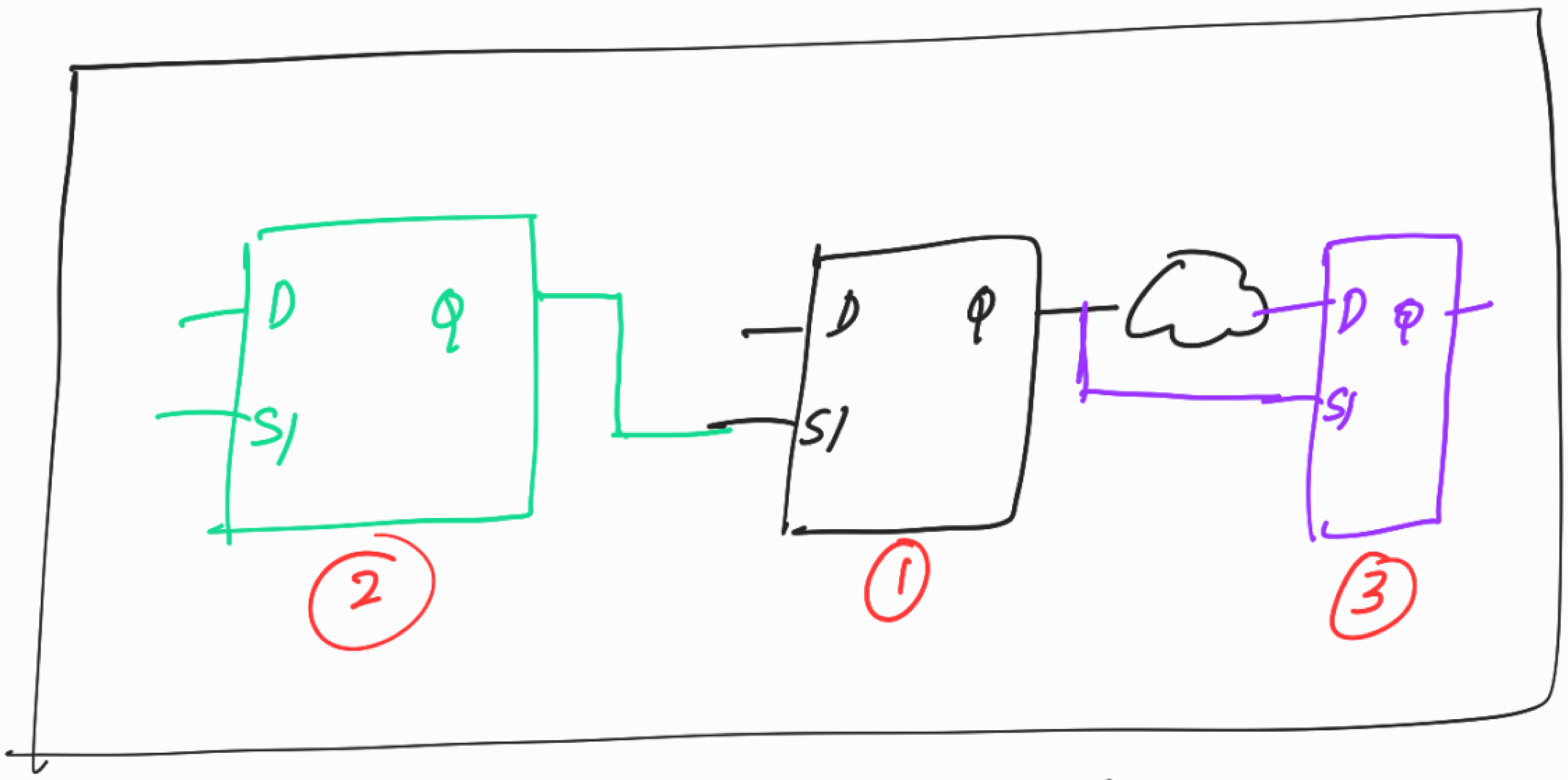
↓ PD team (after placement)



If the original order of scan chain is followed :

→ zig-zag connectivity
→ routing congestion
→ wires will get shorted.

∴ DFT team can give flexibility to the PD team to re-order the scan chains.



After scan chain re-ordering.

NOTE :-

scan insertion
↓
scan compression ⟶ PD team
↓
ATPG
↓
simulation

just done in order to do some cov analysis and improvement.

PD team
↓ after doing scan-chain re-ordering
ATPG
↓
simulation

cov analysis + imp will done here also. But it will not be very time consuming as we have already done it.

original pattern which will be given to the tester for testing the chip after man.

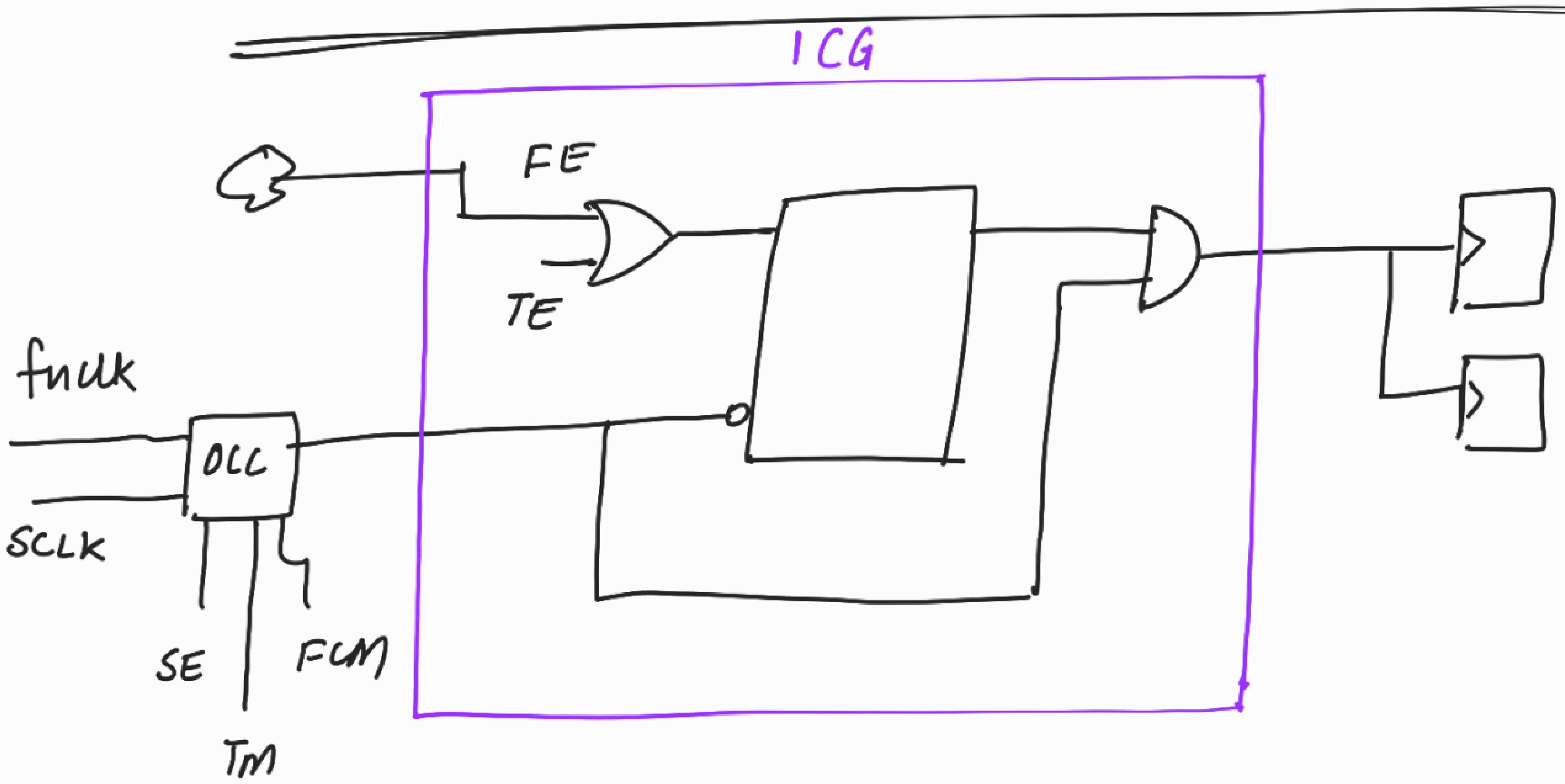
NOTE :- For scan chain re-ordering, PD team need scandef file because that file only contains info about which cells can be re-ordered.

To avoid congestion, PD team has to do scan chain re-order.

RULES OF SCAN CHAIN RE-ORER :-

→ Don't re-order scan cells that have lockup latches, muxes.

HOW TO CONTROL THE TE SIGNAL OF ICG?

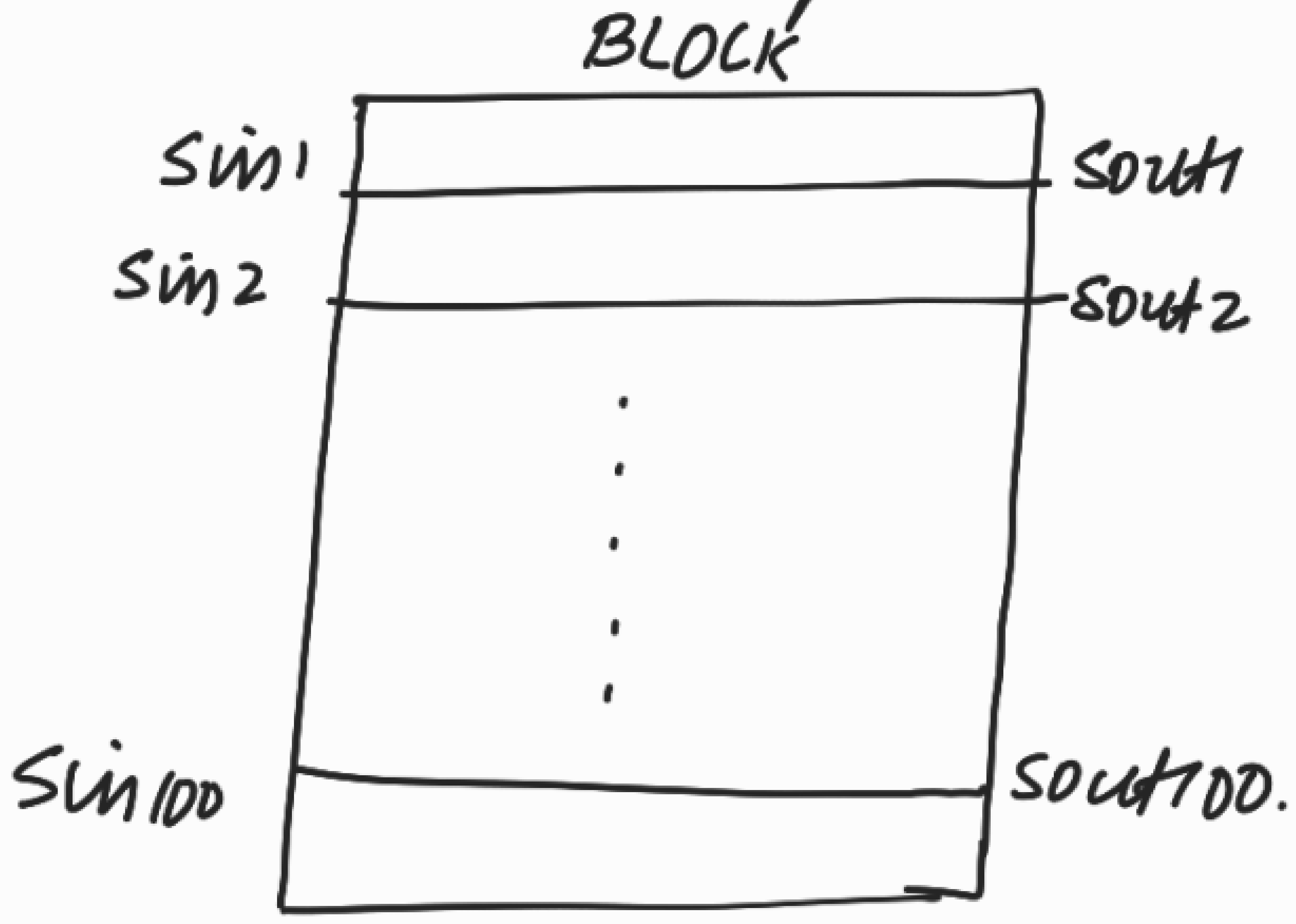


• For fn team, all modules of a block are not operated at the same time. Some modules will operate and some modules will not operate.

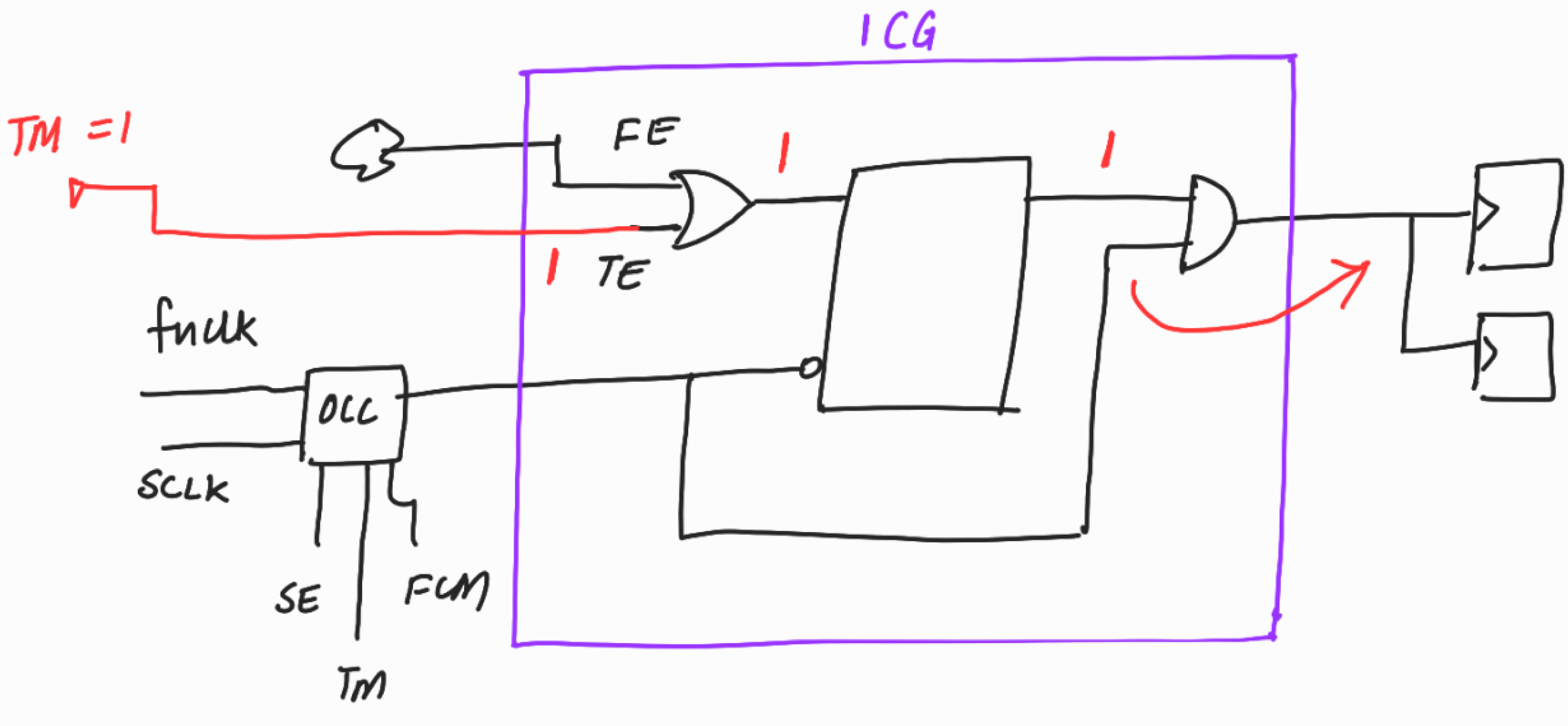
- So fn team will stop the clock of the modules that are not operating at that time inorder to minimize the dynamic power.

- But for DFT team, the complete block will be stitched into scan chains.

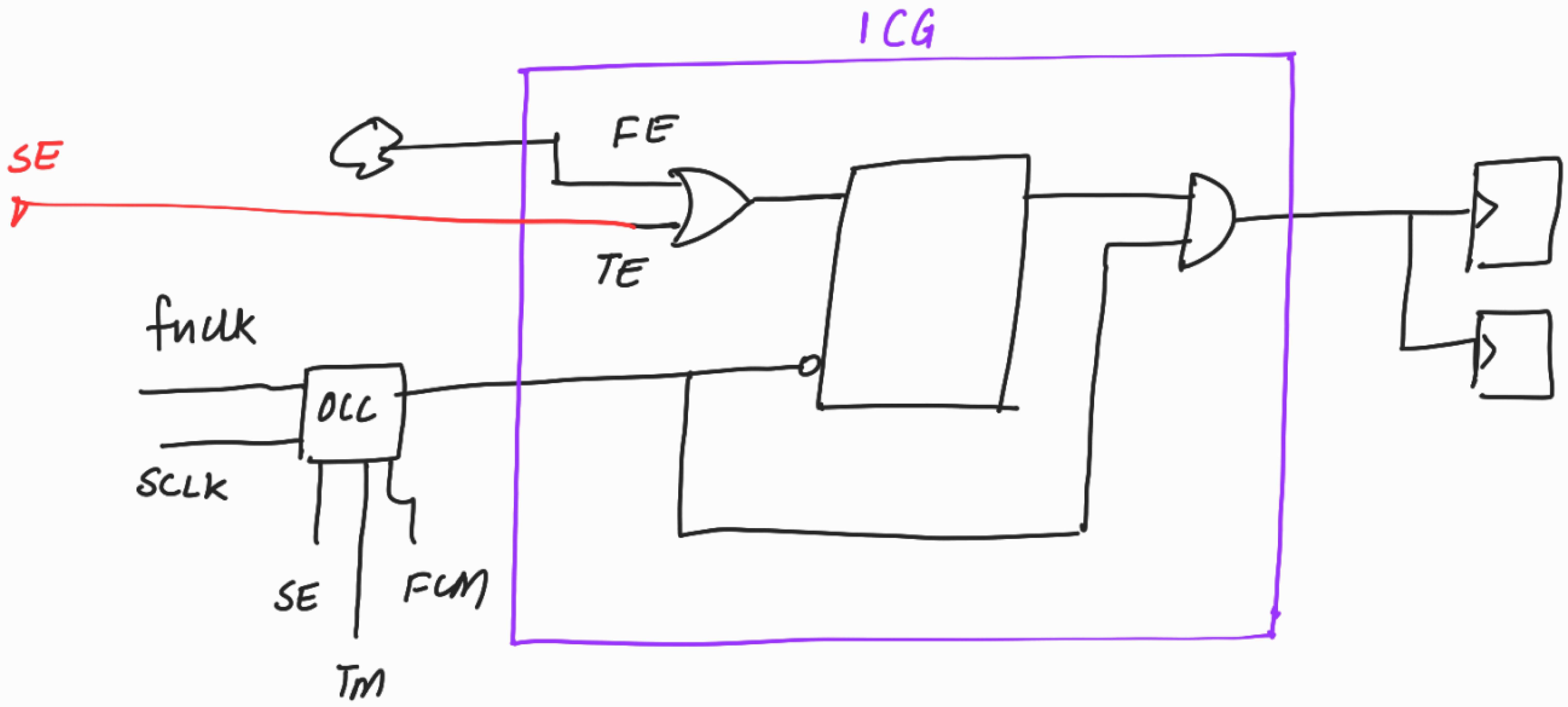
All the scan chains will be operated parallely.



- So during DFT, we have to always propagate clock to all the flops.



Connect TE of ICG to TM.



Connect TE of ICG to SE
(more advantageous)
(Discussed in detail during ATPG class)